

Locality, Non-Locality, and Contextuality

Bell, Kochen-Specker, Bohmian mechanics, and the caution against slogan physics

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Locality, non-locality, and contextuality are delicate concepts in physics and philosophy of quantum mechanics. In this theory they do not prove that everything is relative, that reality is consciousness, or that context simply creates truth. That would be wrong. They are used only to stress that the question “what is the thing” may depend on the measurement frame and relations in which the thing appears.

Bell marks limits for certain families of local hidden-variable explanations of quantum results. Kochen-Specker concerns the difficulty of assigning noncontextual values to all observables while reproducing quantum predictions. Bohmian mechanics is an example of a framework that preserves a certain world-picture at the price of non-locality. These are complex and must not be flattened.

Responsible quantum use = defined concept × distinction between measurement and morality × stated limits × no fast generalization × connection to method / slogan × mysticism × “everything is possible” × mixing consciousness and physics. The aim is not to prove the theory through quantum mechanics, but to prevent the theory from speaking confidently beyond what it knows.

The structural value is contextual realization. Relative potential does not mean there is no truth; it means possibility appears within conditions. A learner understands inside a language and question. Legal evidence gains meaning inside procedure. A medical symptom gains meaning inside body, history, and instrument. A work of art gains meaning inside tradition and reader. This is not relativism; it is responsibility for conditions of appearance.

The link to AI is direct: a model does not generate an answer in a vacuum. Output depends on training, prompt, context, policy, data, interface, and use. Therefore “the model said” is not a source of responsibility. One must ask: in what context, from what information, for what purpose, under what boundary, and who carries responsibility?

Locality and contextuality link to the recursive edge because every answer depends on conditions that invite further testing. They link to science as boundary discipline because they force us not to use quantum terms to escape the hard work of distinction.

Source discipline

Working sources and caution: this chapter relies on the theory itself and on background sources checked outside the prompt: UNESCO on artificial intelligence in education, WHO on ethics and governance of AI for health, the NIST AI RMF for AI risk management, CERN/ATLAS on the Standard Model and the Higgs, NASA and ESA/Planck on cosmology, the Stanford Encyclopedia of Philosophy on Bell, Kochen-Specker and contextuality, and Britannica/ISFDB for the literary identification of Harlan Ellison and I Have No Mouth, and I Must Scream. These sources are not treated as proofs of the theory; they are safeguards against empty borrowing of terms.